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[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.model_selection import cross_val_score

from sklearn.preprocessing import LabelEncoder

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score
```

```
[8]: df = pd.read_csv(r"C:\Users\rutvij kudtarkar\OneDrive\Desktop\titanic.csv")
df.head()
```

```
[8]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

```
[9]: df.isnull().sum()
```

```
[9]: PassengerId      0
Survived           0
Pclass             0
Name               0
Sex                0
Age              177
SibSp              0
Parch              0
Ticket             0
Fare               0
Cabin            687
Embarked           2
dtype: int64
```

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[11]: df['Age'] = df['Age'].fillna(df['Age'].mean())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

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[12]: le = LabelEncoder()

df['Sex'] = le.fit_transform(df['Sex'])
df['Embarked'] = le.fit_transform(df['Embarked'])
```

```
[13]: df.drop(['Name', 'Ticket', 'Cabin'], axis=1, inplace=True)
```

```
[14]: X = df.drop('Survived', axis=1)
y = df['Survived']
```

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[15]: X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
[16]: lr = LogisticRegression(max_iter=1000)
lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)
```

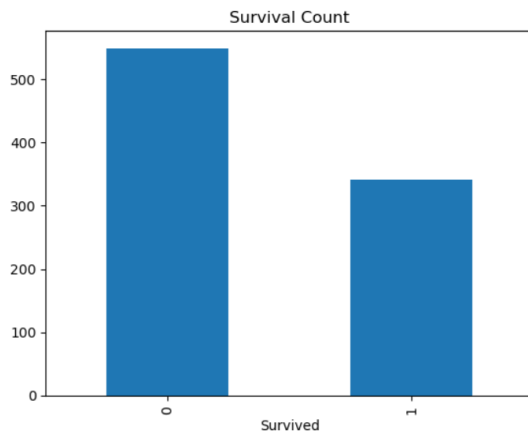
```
[17]: rf = RandomForestClassifier()
rf.fit(X_train, y_train)

y_pred_rf = rf.predict(X_test)
```

```
[18]: print("Accuracy:", accuracy_score(y_test, y_pred_rf))
print("Precision:", precision_score(y_test, y_pred_rf))
print("Recall:", recall_score(y_test, y_pred_rf))
print("F1 Score:", f1_score(y_test, y_pred_rf))
print("ROC AUC:", roc_auc_score(y_test, y_pred_rf))
```

Accuracy: 0.8268156424581006
Precision: 0.8115942028985508
Recall: 0.7567567567567568
F1 Score: 0.7832167832167832
ROC AUC: 0.8164736164736165

```
[20]: df['Survived'].value_counts().plot(kind='bar')  
plt.title("Survival Count")  
plt.show()
```



```
[21]: df['Age'].hist()  
plt.title("Age Distribution")  
plt.show()
```

